Math 10C — Unit 1: Assignments

The assignment from every lesson in the booklet. Printed work is not tracked on the website.

Lesson 1 Assignment *Show your work.*

Part A — Factors, Primes and Composites

1. State the factors of the following.
(a) 19 (b) 34 (c) 45 (d) 48

2. In each case, determine the number of factors of the given whole number.
(a) 9 (b) 13 (c) 29 (d) 35 (e) 54

3. State the numbers in Question 2 which are
(a) prime (b) composite

4. Classify the following whole numbers as prime or composite.
(a) 38 (b) 53 (c) 55 (d) 65

(e) 71 (f) 93 (g) 115 (h) 161

5. Twin primes are defined to be consecutive odd numbers that are both prime (e.g. 3 and 5). List the five other twin prime pairs less than 50.

Part B — Prime Factors and Prime Factorization

6. Answer each part for the number 50.
(a) State the factors of 50. (b) State the prime factors of 50. (c) Express 50 as a product of prime factors.

7. State the prime factors of
(a) 22 (b) 40 (c) 63 (d) 84

8. Explain why the numbers 0 and 1 have no prime factors.

9. Use a division ladder to determine the prime factorization of
(a) 132 (b) 315 (c) 2 431 (d) 2 975

10. Use a factor tree to determine the prime factorization of the following.
(a) 264 (b) 425

(c) 4 199 (d) 11 830

11. In each case, write the number as a product of prime factors.
(a) 175 (b) 847 (c) 3 927 (d) 8 250

Part C — Multiple Choice and Numerical Response

12. *(Multiple Choice)* Which of the following numbers is *not* a prime factor of 3 315? A. 3 B. 5 C. 13 D. 19

13. *(Multiple Choice)* How many of the numbers in the list $2, 3, 5, 13$ are *not* prime factors of 2 310?
A. 4 B. 3 C. 2 D. 1

14. *(Numerical Response)* The sum of all the prime factors of 30 030 is _____.

Record your answer left to right. | | | | |

15. *(Numerical Response)* There is only one set of *prime triplets*: three consecutive odd numbers which are all prime. If the

prime triplets are a, b and c, then the value of abc is _____.

Record your answer left to right. | | | | |

16. *(Numerical Response)* The number 3 993 can be expressed as a product of prime factors in the form $p \times q^r$. The value of $p + q + r$ is _____.

Record your answer left to right. | | | | |

Lesson 2 Assignment *Show your work.*

Part A — Greatest Common Factor

1. State the greatest common factor of each group.
(a) 15 and 24 (b) 24 and 36 (c) 18, 24, and 60

2. Use prime factorization to determine the greatest common factor of
(a) 168 and 210 (b) 135 and 189 (c) 220 and 308

3. Use prime factorization to determine the greatest common factor of
(a) 391 and 493 (b) 216 and 588 (c) 1 350 and 1 575

(d) 574 and 861 (e) 91 and 195 (f) 735 and 1 176

4. Determine the greatest common factor of
(a) 72, 108, and 300 (b) 84, 140, 168, and 210

Part B — Least Common Multiple

5. State the least common multiple of
(a) 6 and 8 (b) 4 and 10 (c) 8 and 12 (d) 20, 30, and 45

6. Use prime factorization to determine the least common multiple of
(a) 15 and 40 (b) 24 and 90 (c) 14 and 231

(d) 22 and 121 (e) 65 and 143 (f) 90 and 315

(g) 175 and 245 (h) 51 and 68 (i) 66 and 156

7. Determine the least common multiple of
(a) 8, 12, and 30 (b) 15, 40, and 84 (c) 5, 7, 11, and 17 (d) 4, 15, 55, and 121

Part C — Multiple Choice, Numerical Response, and Application

8. *(Multiple Choice)* The greatest common factor of 495 and 660 is A. 11 B. 15 C. 33 D. 165

9. *(Multiple Choice)* The greatest common factor of two whole numbers x and y is 14. Which of the statements A, B, C, or D below is false? Answer E if none of the statements is false.
A. x and y must both be even numbers.
B. The product xy must be divisible by 196.
C. x and y are both divisible by 7.
D. Neither x nor y can be a prime number.

10. *(Numerical Response)* The least common multiple of 33, 40, and 44 is _____.

Record your answer left to right. | | | | |

11. A picture book has 725 pages. Every 9th page, starting with page 9, has a red border, and every 12th page, starting with page 12, has a blue border. How many pages have both a red border and a blue border?

Part D — Problem Solving

12. A rectangular floor measures 252 cm by 360 cm. It will be covered with identical square tiles, with no tiles cut. **(a)** What is the largest possible side length of a tile? **(b)** How many of these tiles are needed?

13. Two city buses leave the same stop at 7:00 a.m. Route A returns to the stop every 18 minutes and Route B every 24 minutes. When is the next time both buses are at the stop together?

Lesson 3 Assignment *Show your work.*

Part A — Perfect Squares

1. In each case use prime factorization to determine whether the number is a perfect square. If it is a perfect square, state its square root. Verify your answer using a calculator.
(a) 324 (b) 4 900 (c) 686 (d) 1 050

Part B — Perfect Cubes

2. Consider the number 91 125.
(a) Use a calculator to find the cube root of 91 125.

(b) **Explain** how the prime factorization of 91 125 shows that it is a perfect cube, and use that factorization to verify your calculator answer.

3. In each case use prime factorization to determine whether the number is a perfect cube. Verify your answer using a calculator to find the cube root.
(a) 512 (b) 6 859 (c) 1 024 (d) 3 025

Part C — Square, Cube, Both, or Neither?

4. **(a) Explain** how you could use prime factorization to determine whether a whole number is *both* a perfect square and a perfect cube. **(b)** Find the two smallest whole numbers greater than 1 that are both a perfect square and a perfect cube.

5. Decide whether each number is a perfect square, a perfect cube, both, or neither. State each root that is a whole number.
(a) 256 (b) 1 331 (c) 15 625 (d) 1 800

Part D — Making Perfect Squares and Perfect Cubes

6. Find the smallest whole number each must be multiplied by to give a perfect square, and state the square root of the result.
(a) 18 (b) 245 (c) 600

7. Find the smallest whole number each must be multiplied by to give a perfect cube, and state the cube root of the result.
(a) 32 (b) 1 323 (c) 500

Part E — Multiple Choice and Numerical Response

8. *(Multiple Choice)* The prime factorization of a whole number is $2^a \times 3^6$. For the number to be a perfect square but *not* a perfect cube, the value of a could be A. 3 B. 4 C. 6 D. 9

9. *(Numerical Response)* The smallest whole number that 2 940 must be multiplied by to give a perfect square is _____.

Record your answer left to right. | | | | |

Part F — Problem Solving

10. **(a)** Sam has 20 square tiles and Jada has 49. Who can arrange *all* of their tiles into one solid square? Use prime factorization to explain. **(b)** Can 64 sugar cubes be stacked into one solid cube? Can 72? Use prime factorization to explain.

11. A square garden has an area of 1 764 m^2. **(a)** Use prime factorization to find the side length of the garden. **(b)** How much fencing is needed to enclose it?

12. A cube-shaped storage box has a volume of 2 744 cm^3. Use prime factorization to find the length of each edge.

13. **(a)** Write 28 as a product of prime factors in exponent form. **(b)** Without multiplying anything out, write 28^2 and 28^3 in exponent form. **(c) Explain** why the square of *any* whole number has only even exponents in its prime factorization.

14. Work from the exponents. **(a)** A square has an area of $2^4 \times 3^2 \times 5^2$ cm^2. Find its side length in exponent form and as a whole number. **(b)** A cube has a volume of $2^9 \times 3^6$ cm^3. Find its edge length in exponent form and as a whole number.

Lesson 4 Assignment *Show your work.*

1. State whether the decimal equivalent of each number is repeating or non-repeating. Also state whether it is terminating or non-terminating.

(a) $\frac{3}{4}$ (b) $0.234\,234\,234\ldots$ (c) $-\frac{5}{11}$ (d) $\sqrt{\frac{25}{36}}$

(e) $-\sqrt{23}$ (f) $\sqrt{0.49}$ (g) $-5\frac{3}{8}$ (h) π

2. Classify each statement as true or false.

(a) Every terminating decimal can be expressed as a fraction.

(b) A decimal that repeats forever can never be written as a fraction.

(c) The only decimals that can be written as fractions are ones that terminate.

(d) Every rational number is either a terminating decimal or a repeating decimal.

(e) A single decimal number can be both repeating and non-repeating at the same time.

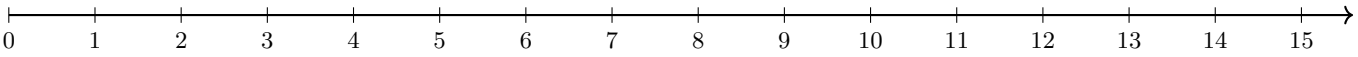
(f) The number π is irrational.

3. Determine whether each number is rational or irrational. Explain your reasoning.

(a) $-\frac{9}{4}$ (b) $0.\overline{567}$ (c) $\sqrt{169}$ (d) $0.3455\,34555\,345555\,3\ldots$

4. Place the following irrational numbers, in order, on the number line below.

(a) $\sqrt{21}$ (b) $\sqrt{0.3}$ (c) $\sqrt{50}$ (d) $2\sqrt{40}$ (e) $3\sqrt{15}$ (f) $\sqrt{7}$



5. Identify each number as rational or irrational. Use a calculator to convert the rational numbers to fractions in simplest form.

(a) 0.6 (b) $\sqrt{\frac{1}{16}}$ (c) $\sqrt{0.0081}$ (d) $-\sqrt{83}$ (e) $-0.\overline{7}$

(f) $-\sqrt{1\frac{11}{25}}$ (g) $5.201\,201\,201\ldots$ (h) $6.835\,8355\,83555\,83\ldots$ (i) $\sqrt{\sqrt{\frac{16}{625}}}$

6. Use a calculator to convert each repeating decimal to an improper fraction in simplest form.

(a) $0.\overline{4}$ (b) $0.\overline{27}$ (c) $2.1\overline{6}$ (d) $0.\overline{312}$ (e) $-1.2\overline{45}$

7. *(Extension)* Use the algebraic procedure to convert each of the following to a fraction in simplest form.
(a) $0.\overline{5}$ (b) $0.\overline{47}$ (c) $0.7\overline{3}$

8. *(Extension)* Use the algebraic procedure to convert each of the following to an improper fraction in simplest form.
(a) $3.0\overline{07}$ (b) $-2.\overline{145}$ (c) $5.83\overline{6}$

9. *(Multiple Choice)* The decimal number representing $\frac{7}{12}$ is
A. terminating and repeating B. terminating and non-repeating
C. non-terminating and repeating D. non-terminating and non-repeating

10. *(Multiple Choice)* Which of the following is an irrational number? A. $\sqrt{196}$ B. $\sqrt{0.045}$ C. $\frac{7}{9}$ D. 2.71

11. *(Multiple Choice)* $7.\overline{9}$ is equal to A. $\frac{79}{10}$ B. $\frac{799}{100}$ C. 7.9 D. 8

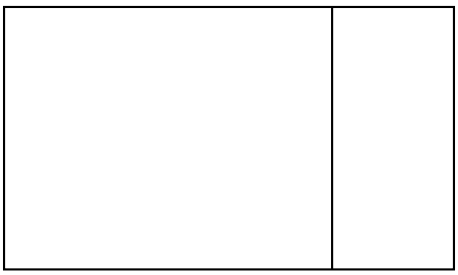
12. *(Numerical Response)* When the repeating decimal $0.3\overline{68}$ is converted to a rational number in simplest form $\frac{a}{b}$, the value of $b - a$ is _____.

Record your answer left to right. | | | | |

Lesson 5 Assignment *Show your work.*

Part A — The Real Number System

1. Complete the Venn diagram below so that it shows how the number sets in this lesson are related: the rectangle is R, the vertical line separates the rationals from the irrationals, so draw and label nested ellipses for Q, I, W, N inside the left region and label $\overline{Q}$ in the right region.



2. For each number below, list all the sets $(N, W, I, Q, \overline{Q}, R)$ to which it belongs, from largest to smallest.

(a) -5 (b) $\sqrt{49}$ (c) $1.414\,213\,88$ (d) $-\frac{9}{4}$

(e) 0 (f) $\sqrt{11}$ (g) $-4.317\,251\,9\ldots$ (h) e

3. Explain why -6 belongs to more number sets than $-\frac{6}{5}$.

4. Use check marks to indicate all the sets to which each number belongs.

	N	W	I	Q	$\overline{Q}$	R
(a) $\frac{1}{6}$						
(b) $456\,921$						
(c) -3						
(d) $4.2\overline{71}$						
(e) 12.5						
(f) $\sqrt{50}$						

	N	W	I	Q	$\overline{Q}$	R
(g) $-\pi$						
(h) $-\frac{22}{7}$						
(i) $-\sqrt{64}$						
(j) $0.000\,03$						
(k) $5.363\,738\ldots$						
(l) $\sqrt{0.25}$						

5. Find one number that satisfies each condition.

(a) An integer, but not a whole number. (b) A rational number, but not an integer. (c) A real number, but not a rational number. (d) A whole number, but not a natural number.

6. Complete each statement with *always*, *sometimes*, or *never*.

(a) A whole number is _____ a natural number. (b) The quotient of two integers is _____ an integer.

(c) A whole number is _____ a rational number. (d) The difference of two integers is _____ an integer.

(e) The square root of a number is _____ in the set $\overline{Q}$.

(f) A negative number is _____ in the set W.

(g) A number in set N is _____ a number in set R.

7. Decide whether each statement is true (T) or false (F).

(a) ___ All natural numbers are integers.

(b) ___ Real numbers consist of rational numbers together with irrational numbers.

(c) ___ The set of integers is nested within the set of rational numbers.

(d) ___ All integers are rational numbers.

(e) ___ All irrational numbers are real numbers.

(f) ___ The set R is nested within the set N.

(g) ___ The set Q is nested within the set W.

(h) ___ There is exactly one number in set W that is not also in set N.

8. Decide whether each statement is true or false.

(a) ___ Every positive number has two square roots, but only one cube root.

(b) ___ Every negative number has one cube root, but no square roots.

Part B — Estimating Square Roots

9. Use estimates to explain why each statement is true.

(a) $\sqrt{11} + \sqrt{14} \neq \sqrt{25}$

(b) $\sqrt{5} + \sqrt{6} + \sqrt{7} \neq \sqrt{18}$

10. Decide whether each statement is true or false.

(a) $\sqrt{16} + \sqrt{4} = \sqrt{16 + 4}$ (b) $\sqrt{16} - \sqrt{4} = \sqrt{16 - 4}$ (c) $\sqrt{16} \times \sqrt{4} = \sqrt{16 \times 4}$ (d) $\sqrt{16} \div \sqrt{4} = \sqrt{16 \div 4}$

11. In each of the following: (i) estimate the value mentally; (ii) use a calculator to find the decimal approximation to the nearest tenth, and decide if the estimate was reasonable.

(a) $\sqrt{23}$ (b) $\sqrt{31.5}$ (c) $5\sqrt{40} - 2\sqrt{90}$ (d) $\frac{2}{3}\sqrt{17.9} + \frac{1}{4}\sqrt{5}$

(e) $\sqrt{137}$ (f) $\sqrt{\sqrt{65}}$ (g) $\sqrt{\sqrt{15} + \sqrt{30.9}}$ (h) $\sqrt{\sqrt{4\,901}}$

12. Estimate each square root to one significant digit.

(a) $\sqrt{542}$ (b) $\sqrt{6\,234}$ (c) $\sqrt{15\,678}$ (d) $\sqrt{715\,000}$

(e) $\sqrt{0.615}$ (f) $\sqrt{0.083\,2}$ (g) $\sqrt{0.008\,12}$ (h) $\sqrt{0.000\,412\,3}$

Part C — Cube Roots and Ordering

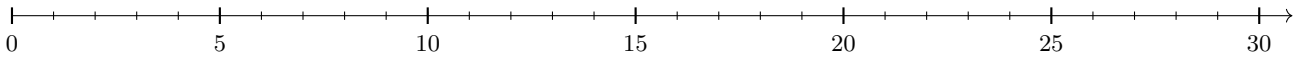
13. In each of the following: (i) estimate the value mentally; (ii) use a calculator to find the decimal approximation to the nearest tenth, and decide if the estimate was reasonable.

(a) $\sqrt[3]{33}$ (b) $\sqrt[3]{4}$ (c) $\sqrt[3]{275}$ (d) $\sqrt[3]{888.8}$

(e) $3\sqrt[3]{67.5} - 2\sqrt[3]{9.3}$ (f) $\frac{3}{5}\sqrt{45} - \frac{1}{3}\sqrt[3]{70}$ (g) $\sqrt[3]{4\sqrt{15}}$

14. Order the following numbers on the number line below, from 0 to 30.

$\sqrt{60}, \quad \sqrt[3]{60}, \quad 4\sqrt{15}, \quad 4\sqrt[3]{15}, \quad 8\sqrt{6}, \quad 8\sqrt[3]{6}$



Part D — Multiple Choice and Numerical Response

15. *(Multiple Choice)* Consider the following statements.

i) The set of integers is nested within the set of natural numbers.
ii) The set of whole numbers is nested within the set of natural numbers.
iii) The set of rational numbers is nested within the set of integers.
iv) The set of irrational numbers is nested within the set of real numbers.

Which of the above statements is false? **A.** i), ii), and iii) only **B.** i), ii), and iv) only **C.** ii), iii), and iv) only Answer **D** if all four statements are false.

16. *(MC)* How many of the numbers $-\sqrt{10}$, $\sqrt{-10}$, $-\sqrt[3]{10}$, $\sqrt[3]{-10}$ do *not* belong to the real number system?

A. 0 **B.** 1 **C.** 2 **D.** 3

17. *(MC)* How many of the numbers $\sqrt{81}$, $\sqrt{8.1}$, $\sqrt{0.81}$, $\sqrt{\frac{9}{16}}$ can be expressed in the form $\frac{a}{b}$, where $a, b \in N$?

A. 1 **B.** 2 **C.** 3 **D.** 4

18. *(Numerical Response)* To the nearest hundredth, the value of $4\sqrt[3]{11}$ is _____.

Record your answer left to right. | | | | |

Part E — Extension: Absolute Value

19. Evaluate each of the following.

(a) $|-6|$ (b) $|15|$ (c) $|4-11|$ (d) $|4|-|11|$

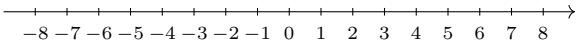
(e) $\big||4|-|11|\big|$ (f) $-|\sqrt[3]{64}|$ (g) $|-\sqrt[3]{64}|$ (h) $|\sqrt[3]{-64}|$

20. Decide whether each statement is true or false.

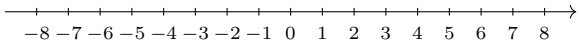
(a) $|x| = x$ if $x > 0$ (b) $|x| = -x$ if $x < 0$

21. In each case, draw a graph to represent the absolute value inequality. The variables are defined on the set of real numbers.

(a) $|x| < 6$



(b) $|a| \geq 4$



Lesson 6 Assignment *Show your work.*

Part A — Evaluating Radicals

1. Mentally evaluate, where possible, using the real number system.

(a) $\sqrt{144}$ (b) $\sqrt[4]{625}$ (c) $4\sqrt[3]{64}$ (d) $\sqrt[5]{7\,776}$

(e) $\sqrt{\frac{9}{49}}$ (f) $\sqrt[4]{\frac{1}{81}}$ (g) $5\sqrt{\frac{1}{25}}$ (h) $-\sqrt{9}$

(i) $\sqrt{-4}$ (j) $\sqrt[5]{-32}$ (k) $4\sqrt[3]{-27}$ (l) $\sqrt[4]{-\frac{1}{81}}$

(m) $5\sqrt{225}$ (n) $\frac{1}{3}\sqrt[5]{243}$ (o) $-\sqrt[9]{-1}$ (p) $\sqrt[3]{-\frac{27}{64}}$

2. State whether the following are true or false.

(a) The square roots of 64 are ± 8. (b) $\sqrt{64} = \pm 8$ (c) If $x^2 = 64,\ x \in R$, then $x = \pm 8$.

3. Use a calculator to evaluate.

(a) $\sqrt[4]{14\,641}$ (b) $\sqrt[5]{-1\,024}$ (c) $-\sqrt[4]{38\,416}$

(d) $-\sqrt[3]{1\,331}$ (e) $\sqrt[3]{-3\,375}$ (f) $-6\sqrt[4]{\frac{1}{1\,296}}$

(g) $5\sqrt[6]{0.000064}$ (h) $\sqrt[4]{-38\,416}$ (i) $\frac{5}{3}\sqrt[4]{\frac{81}{625}}$

4. Evaluate to the nearest hundredth. 5. Evaluate to the nearest tenth.

4. (a) $\sqrt{15}$ 4. (b) $\sqrt[8]{50}$ 4. (c) $-\frac{4}{3}\sqrt[7]{-500}$

5. (a) $\sqrt[5]{-40}$ 5. (b) $-4\sqrt[4]{400}$ 5. (c) $\frac{2}{3}\sqrt[3]{-90}$

Part B — Radical Vocabulary

6. Identify the index and the radicand in each of the following.

(a) $\sqrt[5]{60}$ (b) $\sqrt[6]{81}$ (c) $7\sqrt{23}$

7. Explain the meaning of the index 6 in the radical $\sqrt[6]{81}$.

Part C — Products and Quotients of Radicals

8. Determine whether each statement is true or false.

(a) $\sqrt{56} = \sqrt{7}\sqrt{8}$ (b) $\sqrt{9-4} = \sqrt{9} - \sqrt{4}$ (c) $\sqrt{4} = \dfrac{\sqrt{60}}{\sqrt{15}}$ (d) $\dfrac{\sqrt{36}}{\sqrt{9}} = \sqrt{9}$

(e) $\sqrt{3} + \sqrt{3} = \sqrt{6}$ (f) $\sqrt{3} \times \sqrt{3} = \sqrt{9}$ (g) $\sqrt{\dfrac{1}{3} \times 27} = \sqrt{9}$ (h) $\dfrac{1}{3}\sqrt{27} = \sqrt{9}$

9. Write the following as a single radical in the form $\sqrt{x}$.

(a) $\sqrt{6} \times \sqrt{9}$ (b) $\sqrt{15} \times \sqrt{3}$ (c) $\sqrt{4} \times \sqrt{11}$ (d) $\sqrt{8} \times \sqrt{9}$

(e) $\dfrac{\sqrt{27}}{\sqrt{3}}$ (f) $\dfrac{\sqrt{49}}{\sqrt{7}}$ (g) $\dfrac{\sqrt{14}\sqrt{6}}{\sqrt{3}}$ (h) $\dfrac{\sqrt{\sqrt{64}}}{\sqrt{\sqrt{16}}}$

10. Express as a product of two radicals.

(a) $\sqrt{77}$ (b) $\sqrt{51}$ (c) $\sqrt{95}$ (d) $\sqrt{121}$

Part D — Multiple Choice and Numerical Response

11. *(Multiple Choice)* Consider the following statements.

I. The cube roots of -64 are ± 4.
II. The fourth roots of 625 are ± 5.
III. $-\sqrt[3]{512} = \sqrt[3]{-512}$
IV. $-\sqrt[4]{81} = \sqrt[4]{-81}$

Which of the statements above are true?

A. II and III only **B.** I, II and III only **C.** I, II, III, and IV **D.** some other combination of I, II, III, and IV

12. *(Multiple Choice)* In the radical $\sqrt{23}$,

A. the index is 2 and the radicand is $\sqrt{23}$ **B.** the index is 1 and the radicand is 1
C. the index is 23 and the radicand is 1 **D.** the index is 2 and the radicand is 23

13. *(Numerical Response)* To the nearest hundredth, the value of $\sqrt[5]{-\dfrac{5}{6}} + 3\sqrt[4]{\dfrac{5}{6}}$ is _____.

Record your answer left to right. | | | | |

Lesson 7 Assignment *Show your work.*

Part A — Entire and Mixed Radicals (Square Roots)

1. State whether each radical is written as a mixed radical or an entire radical.
(a) $\sqrt{42}$ (b) $5\sqrt{3}$ (c) $\sqrt{100}$ (d) $0.7\sqrt{11}$

2. Convert the following radicals to mixed radicals in simplest form.
(a) $\sqrt{12}$ (b) $\sqrt{28}$ (c) $\sqrt{48}$ (d) $\sqrt{99}$

(e) $4\sqrt{27}$ (f) $-6\sqrt{80}$ (g) $3\sqrt{125}$ (h) $-5\sqrt{72}$

3. Convert the following radicals to mixed radicals in simplest form. **Two of them cannot be converted** — identify them and explain why.
(a) $\sqrt{150}$ (b) $\sqrt{392}$ (c) $-\frac{3}{4}\sqrt{128}$ (d) $\frac{1}{9}\sqrt{567}$

(e) $\sqrt{605}$ (f) $5\sqrt{242}$ (g) $\sqrt{3\,125}$ (h) $\sqrt{70}$

(i) $-\frac{7}{12}\sqrt{432}$ (j) $\sqrt{845}$ (k) $3\sqrt{578}$ (l) $-2\sqrt{847}$

(m) $5\sqrt{363}$ (n) $\sqrt{776}$ (o) $\frac{3}{7}\sqrt{1\,372}$ (p) $\frac{11}{13}\sqrt{143}$

4. Convert the following radicals to mixed radicals where the radicand is a whole number.
(a) $\sqrt{\frac{5}{49}}$ (b) $\sqrt{\frac{11}{9}}$ (c) $\sqrt{\frac{32}{49}}$ (d) $6\sqrt{\frac{20}{9}}$

5. Convert the following to entire radical form.
(a) $3\sqrt{5}$ (b) $4\sqrt{6}$ (c) $6\sqrt{11}$ (d) $10\sqrt{3}$

(e) $2\sqrt{49}$ (f) $-7\sqrt{6}$ (g) $8\sqrt{11}$ (h) $-9\sqrt{2}$

6. Convert the following to entire radical form.
(a) $\frac{1}{6}\sqrt{180}$ (b) 18 (c) $\frac{5}{3}\sqrt{18}$ (d) $2^2\sqrt{13}$

Do not use a calculator for questions 7 and 8.

7. Given that $\sqrt{7} \approx 2.65$ and $\sqrt{70} \approx 8.37$, find the approximate value of each radical.

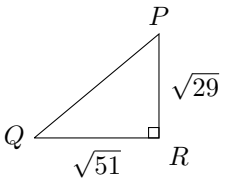
(a) $\sqrt{700}$ (b) $\sqrt{7\,000}$ (c) $\sqrt{700\,000}$ (d) $\sqrt{0.07}$

(e) $\sqrt{0.7}$ (f) $\sqrt{28}$ (g) $\sqrt{630}$ (h) $\sqrt{\frac{7}{16}}$

8. Arrange the following radicals in order from greatest to least. $4\sqrt{5}$, $6\sqrt{2}$, $\sqrt{95}$, $3\sqrt{7}$, $\frac{1}{3}\sqrt{450}$

Part B — Radicals and the Pythagorean Theorem

9. Students are finding the length of PQ in $\triangle PQR$ below. **Louis** expresses each side to the nearest hundredth and then calculates PQ to the nearest hundredth. **Asia** uses the entire radical form for each side and expresses her answer to the nearest hundredth.



(a) Complete Louis's work.

$\sqrt{51} \approx 7.14$, $\sqrt{29} \approx 5.39$

$PQ^2 \approx 7.14^2 + 5.39^2$

$\approx$ _____

$PQ \approx$ _____

Complete Asia's work.

$PQ^2 = \left(\sqrt{51}\right)^2 + \left(\sqrt{29}\right)^2$

$=$ ____ $+$ ____ $=$ ____

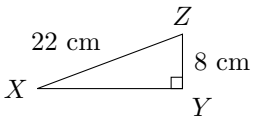
$PQ = \sqrt{\quad} \approx$ _____

(b) Which student's answer is more accurate? Explain.

(c) State the **exact** answer as a mixed radical.

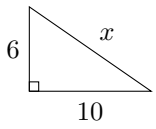
10. Use $c^2 = a^2 + b^2$ in the triangle shown to calculate the length of XY.

(i) an entire radical (ii) a mixed radical (iii) a decimal, to the nearest hundredth

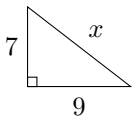


11. Find the lengths of the missing sides. Express the answers in simplest mixed radical form.

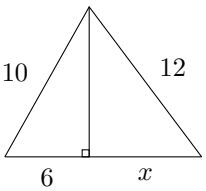
(a)



(b)

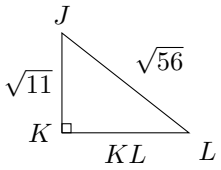


(c)



12. *(Multiple Choice)* The length of KL can be represented by which of the following?

A. $\sqrt{67}$ B. $3\sqrt{5}$ C. $\sqrt{30}$ D. $9\sqrt{2}$



13. *(Multiple Choice)* Without using a calculator, determine which of the following radicals is *not* equal to the others. A. $10\sqrt{3}$ B. $\sqrt{300}$ C. $5\sqrt{12}$ D. $3\sqrt{48}$

Part C — Applications (Numerical Response)

Use the following information to answer question 14. A physics textbook gives the distance, d kilometres, that a person can see to the horizon on a clear day as $d = \sqrt{13h}$, where h is the person's eye-level height above the ground, in metres. Standing on the ground, Maya's eye-level height is 1.4 m above the ground.

14. The distance Maya can see to the horizon while standing on the roof of a building 1 098.6 m high can be written in simplest form as $a\sqrt{b}$. Find the value of $a + b$.

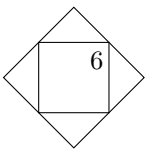
Record your answer left to right. ☐☐☐☐

Use the following information to answer question 15. Heron's Formula, developed in Ancient Greece, gives the area of a triangle as $A = \sqrt{s(s-a)(s-b)(s-c)}$, where $s = \dfrac{a+b+c}{2}$ and a, b, c are the lengths of the three sides.

15. The area of a triangle whose sides measure 10, 13 and 15 can be written in simplest form as $p\sqrt{114}$, where $p \in N$. Find the value of p.

Record your answer left to right. ☐☐☐☐

16. The smaller square shown has side length 6 cm. The side length of the larger square can be written in simplest form as $p\sqrt{q}$, where $p, q \in N$. Find the value of pq.



Record your answer left to right. ☐☐☐☐

Lesson 8 Assignment *Show your work.*

Part A — Radicals With Index 3 and Higher

1. Convert the following radicals to mixed radicals in simplest form.

(a) $\sqrt[3]{135}$ (b) $\sqrt[3]{375}$ (c) $\sqrt[3]{4000}$ (d) $4\sqrt[3]{-192}$

(e) $\frac{4}{9}\sqrt[3]{250}$ (f) $3\sqrt[4]{512}$ (g) $\sqrt[5]{-320}$ (h) $-3\sqrt[3]{1029}$

2. Convert the following mixed radicals to entire radicals.

(a) $3\sqrt[4]{5}$ (b) $4\sqrt[3]{3}$ (c) $-2\sqrt[4]{7}$ (d) $-6\sqrt[3]{7}$

(e) $3\sqrt[5]{4}$ (f) $\frac{2}{3}\sqrt[3]{108}$ (g) $\frac{2}{5}\sqrt[4]{50}$ (h) $-4\sqrt[3]{11}$

3. **(No calculator.)** Arrange the following radicals in order from least to greatest. $5\sqrt[6]{1}$, $-4\sqrt[3]{-64}$, $\frac{3}{2}\sqrt[4]{81}$, $2\sqrt{\sqrt[3]{729}}$

4. **(No calculator.)** Consider the radicals $3\sqrt[3]{5}$, $2\sqrt[3]{9}$, $4\sqrt[3]{3}$, $2\sqrt[3]{7}$.

(a) Explain how to arrange the radicals in order from least to greatest without a calculator.

(b) Arrange the radicals in order from least to greatest.

5. *(Multiple Choice)* $\sqrt[3]{648}$ is equivalent to A. $4\sqrt[3]{18}$ B. $2\sqrt[3]{243}$ C. $6\sqrt[3]{3}$ D. $216\sqrt[3]{3}$

6. *(Multiple Choice)* Consider the two statements **Statement 1:** $-2\sqrt[4]{5} = 2\sqrt[4]{-5}$ and **Statement 2:** $-4\sqrt[3]{6} = 4\sqrt[3]{-6}$. Which of the following is correct?

A. Both statements are true. B. Both statements are false. C. Statement 1 is true, Statement 2 is false. D. Statement 1 is false, Statement 2 is true.

7. *(Numerical Response)* The mixed radical $\frac{1}{9}\sqrt[3]{189}$ can be converted to a mixed radical in simplest form $a\sqrt[3]{b}$. The value of $a + b$, to the nearest tenth, is _____.

Record your answer left to right. | | | | |

Part B — Extension: Radicals With Variables

Assume every variable represents a non-negative number.

8. Express each as an entire radical.

(a) $5\sqrt{w}$ (b) $7\sqrt{d^2}$ (c) $6\sqrt{3pt^3}$ (d) $-4\sqrt[3]{x^2}$

(e) $p\sqrt{p}$ (f) $t\sqrt{5m^3}$ (g) $9d^2\sqrt{d^2e}$ (h) $6a^2b\sqrt{2a^3b}$

(i) $3m^2n\sqrt{5}$ (j) $3pq^2\sqrt[3]{4pq}$ (k) $5p^6q^7\sqrt{p^3r}$ (l) $3xy^2\sqrt[4]{4x^3}$

9. Express each as a mixed radical in simplest form.

(a) $\sqrt{a^7}$ (b) $\sqrt{t^5}$ (c) $\sqrt{x^{13}}$ (d) $\sqrt[3]{x^7}$ (e) $\sqrt[3]{c^{11}}$ (f) $\sqrt[4]{z^{10}}$

10. Express each as a mixed radical in simplest form.

(a) $\sqrt{18m^2}$ (b) $\sqrt{36k^5}$ (c) $\sqrt{45m^3n^4}$ (d) $\sqrt{432a^7w^9}$

(e) $3\sqrt{45c^6d^5}$ (f) $-4\sqrt{31p^6q^{10}}$ (g) $5m^2n^3\sqrt{20m^7n^4}$ (h) $\frac{3}{4}m\sqrt{64m^5n^8}$

(i) $9x^5y^{11}\sqrt{200x^7y^6}$ (j) $\sqrt[3]{1029x^{10}}$ (k) $3\sqrt[3]{432y^{16}}$ (l) $\sqrt[4]{162m^{13}}$

11. *(Multiple Choice)* $\sqrt{7z}\cdot\sqrt{3z}$ is equivalent to A. $\sqrt{21z}$ B. $\sqrt{441z^2}$ C. $21\sqrt{z}$ D. $z\sqrt{21}$

Lesson 9 Assignment *Show your work.*

Part A — Multiple Choice

1. The height of a building is $9\sqrt[5]{2\,500}$ metres. In the number $9\sqrt[5]{2\,500}$, the index and radicand are respectively
A. 5 and 2 500 B. 9 and 2 500
C. 5 and 9 D. 2 500 and 5

2. Three statements are given below.
Statement 1: $45 = 9\sqrt{5}$ Statement 2: $\sqrt{75} = 5\sqrt{3}$
Statement 3: $6\sqrt{2} = 72$
Which of the statements above is/are true?
A. 1 only B. 2 only C. 1 and 2 only D. 2 and 3 only

3. Three students were asked to write the radical $\sqrt{7\,938}$ in another form. The answers given were Student I: $63\sqrt{7}$, Student II: $21\sqrt{18}$, Student III: $9\sqrt{98}$. A correct answer was given by
A. only Student III B. only Students II and III
C. all three students D. some other combination of students not given above

4. The area of a circle of radius r is given by the formula $A = \pi r^2$. A circle of radius 8 cm has an area of 64π cm^2. If a circle has an area of 200π cm^2, then the exact length of its radius in cm is
A. 200 B. $10\sqrt{2}$ C. $20\sqrt{2}$ D. $4\sqrt{5}$

5. Consider the following three equations.

$$9\sqrt[3]{2} = \sqrt[3]{x} \qquad 2\sqrt{x} = y\sqrt{2} \qquad 16\sqrt{y} = z\sqrt{6}$$

Which of the statements below is correct? A. $x < y < z$ B. $z < x < y$ C. $y < z < x$ D. $z < y < x$

Part B — Numerical Response

6. *(Numerical Response)* When $9\sqrt[3]{5}$ is written as an entire radical, the value of the radicand is _____.

Record your answer left to right. | | | | |

7. *(Numerical Response)* The volume of a cube of edge length x cm is given by the formula $V = x^3$. A storage cube has a volume of 1 875 cm^3. A student determined that the exact length of each edge of the cube could be written in the form $a\sqrt[3]{b}$, where a and b are whole numbers. The value of $a + b$ is _____.

Record your answer left to right. | | | | |

Part C — Written Response

8. A group of students invented a simple card game based on the real number system. There are 50 cards in the deck, each showing a number, and a card carries points for *every* category it belongs to.

Natural 5 Whole 6 Integer 8 Rational 4 Irrational 12 Non-real 2

For example, a card showing the number 11 earns $5+6+8+4=23$ points, because 11 is a natural number, a whole number, an integer, *and* a rational number.

(a) A student selects a card that has the number -9 on it. How many points are awarded for this card?

(b) Each student is dealt three cards, and the student with the most points wins. Determine the point total for each of the three students below, and state who wins.

Student A: $\frac{5}{6},\ \sqrt{21},\ 0$ Student B: $-7,\ \sqrt{\frac{9}{16}},\ \sqrt{7}$ Student C: $-\sqrt{81},\ \sqrt{-81},\ 81$

The winner is _____.

Answer Key — Lesson Assignments

Lesson 1 Assignment

1. (a) $1, 19$ (b) $1, 2, 17, 34$ (c) $1, 3, 5, 9, 15, 45$ (d) $1, 2, 3, 4, 6, 8, 12, 16, 24, 48$. **2.** (a) 3 (b) 2 (c) 2 (d) 4 (e) 8. **3.** (a) $13, 29$ (b) $9, 35, 54$. **4.** (a) composite (b) prime (c) composite (d) composite (e) prime (f) composite (g) composite (h) composite. **5.** 5 & 7; 11 & 13; 17 & 19; 29 & 31; 41 & 43. **6.** (a) $1, 2, 5, 10, 25, 50$ (b) $2, 5$ (c) $2 \times 5 \times 5 = 2 \times 5^2$. **7.** (a) $2, 11$ (b) $2, 5$ (c) $3, 7$ (d) $2, 3, 7$. **8.** 1 has only one factor, 1, which is not prime; 0 can't be written as a product of primes (multiplying primes never gives 0), so neither has a prime factorization. **9.** (a) $2^2 \times 3 \times 11$ (b) $3^2 \times 5 \times 7$ (c) $11 \times 13 \times 17$ (d) $5^2 \times 7 \times 17$. **10.** (a) $2^3 \times 3 \times 11$ (b) $5^2 \times 17$ (c) $13 \times 17 \times 19$ (d) $2 \times 5 \times 7 \times 13^2$. **11.** (a) $5^2 \times 7$ (b) 7×11^2 (c) $3 \times 7 \times 11 \times 17$ (d) $2 \times 3 \times 5^3 \times 11$. **12.** D. **13.** D. **14.** 41. **15.** 105. **16.** 17.

Lesson 2 Assignment

1. (a) 3 (b) 12 (c) 6. **2.** (a) 42 (b) 27 (c) 44. **3.** (a) 17 (b) 12 (c) 225 (d) 287 (e) 13 (f) 147. **4.** (a) 12 (b) 14. **5.** (a) 24 (b) 20 (c) 24 (d) 180. **6.** (a) 120 (b) 360 (c) 462 (d) 242 (e) 715 (f) 630 (g) 1 225 (h) 204 (i) 1 716. **7.** (a) 120 (b) 840 (c) 6 545 (d) 7 260. **8.** D. **9.** E. **10.** 1 320. **11.** 20. **12.** (a) 36 cm (b) 70 tiles. **13.** 8:12 a.m.

Lesson 3 Assignment

1. (a) yes, 18 (b) yes, 70 (c) no (d) no. **2.** (a) 45 (b) $91\,125 = 3^6 \times 5^3$; both exponents are multiples of 3, so the cube root is $3^2 \times 5 = 45$. **3.** (a) yes, 8 (b) yes, 19 (c) no (d) no. **4.** (a) Factor the number; it is both exactly when every exponent is a multiple of 6. (b) $64 = 2^6$ and $729 = 3^6$. **5.** (a) square only, 16 (b) cube only, 11 (c) both, 125 and 25 (d) neither. **6.** (a) 2; 6 (b) 5; 35 (c) 6; 60. **7.** (a) 2; 4 (b) 7; 21 (c) 2; 10. **8.** B. **9.** 15. **10.** (a) Jada: $49 = 7^2$ is a perfect square; $20 = 2^2 \times 5$ is not. (b) $64 = 2^6$ yes $(4 \times 4 \times 4)$; $72 = 2^3 \times 3^2$ no. **11.** (a) 42 m (b) 168 m. **12.** 14 cm. **13.** (a) $2^2 \times 7$ (b) $2^4 \times 7^2$; $2^6 \times 7^3$ (c) squaring doubles every exponent. **14.** (a) $2^2 \times 3 \times 5 = 60$ cm (b) $2^3 \times 3^2 = 72$ cm.

Lesson 4 Assignment

1. (a) non-repeating, terminating (b) repeating, non-terminating (c) repeating, non-terminating (d) repeating, non-terminating (e) non-repeating, non-terminating (f) non-repeating, terminating (g) non-repeating, terminating (h) non-repeating, non-terminating. **2.** (a) T (b) F (c) F (d) T (e) F (f) T. **3.** (a) rational (terminates, -2.25) (b) rational (repeating) (c) rational ($\sqrt{169} = 13$) (d) irrational (non-repeating, non-terminating). **4.** $\sqrt{0.3} < \sqrt{7} < \sqrt{21} < \sqrt{50} < 3\sqrt{15} < 2\sqrt{40}$. **5.** (a) rational, $\frac{3}{5}$ (b) rational, $\frac{1}{4}$ (c) rational, $\frac{9}{100}$ (d) irrational (e) rational, $-\frac{7}{9}$ (f) rational, $-\frac{6}{5}$ (g) rational, $\frac{1732}{333}$ (h) irrational (i) rational, $\frac{2}{5}$. **6.** (a) $\frac{4}{9}$ (b) $\frac{3}{11}$ (c) $\frac{13}{6}$ (d) $\frac{104}{333}$ (e) $-\frac{137}{110}$. **7.** (a) $\frac{5}{9}$ (b) $\frac{47}{99}$ (c) $\frac{11}{15}$. **8.** (a) $\frac{2977}{990}$ (b) $-\frac{2143}{999}$ (c) $\frac{1751}{300}$. **9.** C. **10.** B. **11.** D. **12.** $\frac{73}{198}$, so $b - a = 125$.

Lesson 5 Assignment

1. Nested ellipses $Q \supset I \supset W \supset N$ in the left region; $\overline{Q}$ in the right region; the rectangle is R. **2.** (a) R, Q, I (b) R, Q, I, W, N (c) R, Q (d) R, Q (e) R, Q, I, W (f) $R, \overline{Q}$ (g) $R, \overline{Q}$ (h) $R, \overline{Q}$. **3.** -6 is an integer, so it is in I, Q, R (three sets); $-\frac{6}{5}$ is not an integer, so it is in Q, R only (two sets). **4.** (a) Q, R (b) N, W, I, Q, R (c) I, Q, R (d) Q, R (e) Q, R (f) $\overline{Q}, R$ (g) $\overline{Q}, R$ (h) Q, R (i) I, Q, R (j) Q, R (k) $\overline{Q}, R$ (l) Q, R. **5.** (answers vary) (a) -4 (b) $\frac{3}{7}$ (c) $\sqrt{2}$ (d) 0. **6.** (a) sometimes (b) sometimes (c) always (d) always (e) sometimes (f) never (g) always. **7.** (a) T (b) T (c) T (d) T (e) T (f) F (g) F (h) T. **8.** (a) true (b) true. **9.** (a) $3 + 4 = 7 \neq 5$ (b) $2 + 2 + 3 = 7 \neq 4$ (mental whole-number estimates). **10.** (a) false (b) false (c) true (d) true. **11.** (a) i) 5 ii) 4.8 (b) i) 6 ii) 5.6 (c) i) 12 ii) 12.6 (d) i) 3.2 ii) 3.4 (e) i) 12 ii) 11.7 (f) i) 3 ii) 2.8 (g) i) 3 ii) 3.1 (h) i) 8 ii) 8.4. **12.** (a) 20 (b) 80 (c) 100 (d) 800 (e) 0.8 (f) 0.3 (g) 0.09 (h) 0.02. **13.** (a) i) 3 ii) 3.2 (b) i) 2 ii) 1.6 (c) i) between 6 and 7 ii) 6.5 (d) i) 10 ii) 9.6 (e) i) 8 ii) 8.0 (f) i) 2.9 ii) 2.7 (g) i) 2 ii) 2.5. **14.** $\sqrt[3]{60} < \sqrt{60} < 4\sqrt[3]{15} < 8\sqrt[3]{6} < 4\sqrt{15} < 8\sqrt{6}$ $(3.9 < 7.7 < 9.9 < 14.5 < 15.5 < 19.6)$. **15.** A **16.** B **17.** C **18.** 8.90. **19.** (a) 6 (b) 15 (c) 7 (d) -7 (e) 7 (f) -4 (g) 4 (h) 4. **20.** (a) true (b) true. **21.** (a) open circles at -6 and 6, shaded between; (b) solid dots at -4 and 4, shaded outward in both directions.

Lesson 6 Assignment

1. (a) 12 (b) 5 (c) 16 (d) 6 (e) $\frac{3}{7}$ (f) $\frac{1}{3}$ (g) 1 (h) -3 (i) not possible (j) -2 (k) -12 (l) not possible (m) 75 (n) 1 (o) 1 (p) $-\frac{3}{4}$. **2.** (a) true (b) false (c) true. **3.** (a) 11 (b) -4 (c) -14 (d) -11 (e) -15 (f) -1 (g) 1 (h) not possible (i) 1. **4.** (a) 3.87 (b) 1.63 (c) 3.24. **5.** (a) -2.1 (b) -17.9 (c) -3.0. **6.** (a) index 5, radicand 60 (b) index 6, radicand 81 (c) index 2, radicand 23. **7.** The index 6 says that $\sqrt[6]{81}$ must be used as a factor 6 times to give a product of 81: $\left(\sqrt[6]{81}\right)^6 = 81$. **8.** (a) true (b) false (c) true (d) false (e) false (f) true (g) true (h) false. **9.** (a) $\sqrt{54}$ (b) $\sqrt{45}$ (c) $\sqrt{44}$ (d) $\sqrt{72}$ (e) $\sqrt{9}$ (f) $\sqrt{7}$ (g) $\sqrt{28}$ (h) $\sqrt{2}$. **10.** (a) $\sqrt{7}\sqrt{11}$ (b) $\sqrt{3}\sqrt{17}$ (c) $\sqrt{5}\sqrt{19}$ (d) $\sqrt{11}\sqrt{11}$. **11.** A **12.** D **13.** 1.90.

Lesson 7 Assignment

1. (a) entire (b) mixed (c) entire (d) mixed. **2.** (a) $2\sqrt{3}$ (b) $2\sqrt{7}$ (c) $4\sqrt{3}$ (d) $3\sqrt{11}$ (e) $12\sqrt{3}$ (f) $-24\sqrt{5}$ (g) $15\sqrt{5}$ (h) $-30\sqrt{2}$. **3.** (a) $5\sqrt{6}$ (b) $14\sqrt{2}$ (c) $-6\sqrt{2}$ (d) $\sqrt{7}$ (e) $11\sqrt{5}$ (f) $55\sqrt{2}$ (g) $25\sqrt{5}$ (h) cannot be converted: $70 = 2 \cdot 5 \cdot 7$ is square-free (i) $-7\sqrt{3}$ (j) $13\sqrt{5}$ (k) $51\sqrt{2}$ (l) $-22\sqrt{7}$ (m) $55\sqrt{3}$ (n) $2\sqrt{194}$ (o) $6\sqrt{7}$ (p) cannot be simplified: $143 = 11 \cdot 13$ is square-free. **4.** (a) $\frac{1}{7}\sqrt{5}$ (b) $\frac{1}{3}\sqrt{11}$ (c) $\frac{4}{7}\sqrt{2}$ (d) $4\sqrt{5}$. **5.** (a) $\sqrt{45}$ (b) $\sqrt{96}$ (c) $\sqrt{396}$ (d) $\sqrt{300}$ (e) $\sqrt{196}$ (f) $-\sqrt{294}$ (g) $\sqrt{704}$ (h) $-\sqrt{162}$. **6.** (a) $\sqrt{5}$ (b) $\sqrt{324}$ (c) $\sqrt{50}$ (d) $\sqrt{208}$. **7.** (a) 26.5 (b) 83.7 (c) 837 (d) 0.265 (e) 0.837 (f) 5.3 (g) 25.11 (h) 0.6625. **8.** $\sqrt{95} > 4\sqrt{5} > 6\sqrt{2} > 3\sqrt{7} > \frac{1}{3}\sqrt{450}$. **9.** (a) Louis 8.95, Asia 8.94 (b) Asia — she used exact values, so she rounded only once (c) $4\sqrt{5}$. **10.** (i) $\sqrt{420}$ cm (ii) $2\sqrt{105}$ cm (iii) 20.49 cm. **11.** (a) $2\sqrt{34}$ (b) $\sqrt{130}$ (c) $4\sqrt{5}$. **12.** B. **13.** D. **14.** 153. **15.** 6. **16.** 12.

Lesson 8 Assignment

1. (a) $3\sqrt[3]{5}$ (b) $5\sqrt[3]{3}$ (c) $10\sqrt[3]{4}$ (d) $-16\sqrt[3]{3}$ (e) $\frac{20}{9}\sqrt[3]{2}$ (f) $12\sqrt[4]{2}$ (g) $-2\sqrt[5]{10}$ (h) $-21\sqrt[3]{3}$. **2.** (a) $\sqrt[4]{405}$ (b) $\sqrt[3]{192}$ (c) $-\sqrt[4]{112}$ (d) $\sqrt[3]{-1512}$ (e) $\sqrt[5]{972}$ (f) $\sqrt[3]{32}$ (g) $\sqrt[4]{\frac{32}{25}}$ (h) $\sqrt[3]{-704}$. **3.** $\frac{3}{2}\sqrt[4]{81} < 5\sqrt[6]{1} < 2\sqrt{\sqrt[3]{729}} < -4\sqrt[3]{-64}$. **4.** (a) convert each to an entire radical of index 3 and compare radicands (b) $2\sqrt[3]{7}$, $2\sqrt[3]{9}$, $3\sqrt[3]{5}$, $4\sqrt[3]{3}$. **5.** C. **6.** D. **7.** 7.3. **8.** (a) $\sqrt{25w}$ (b) $\sqrt{49d^2}$ (c) $\sqrt{108pt^3}$ (d) $\sqrt[3]{-64x^2}$ (e) $\sqrt{p^3}$ (f) $\sqrt{5m^3t^2}$ (g) $\sqrt{81d^6e}$ (h) $\sqrt{72a^7b^3}$ (i) $\sqrt{45m^4n^2}$ (j) $\sqrt[3]{108p^4q^7}$ (k) $\sqrt{25p^{15}q^{14}r}$ (l) $\sqrt[4]{324x^7y^8}$. **9.** (a) $a^3\sqrt{a}$ (b) $t^2\sqrt{t}$ (c) $x^6\sqrt{x}$ (d) $x^2\sqrt[3]{x}$ (e) $c^3\sqrt[3]{c^2}$ (f) $z^2\sqrt[4]{z^2}$. **10.** (a) $3m\sqrt{2}$ (b) $6k^2\sqrt{k}$ (c) $3mn^2\sqrt{5m}$ (d) $12a^3w^4\sqrt{3aw}$ (e) $9c^3d^2\sqrt{5d}$ (f) $-4p^3q^5\sqrt{31}$ (g) $10m^5n^5\sqrt{5m}$ (h) $6m^3n^4\sqrt{m}$ (i) $90x^8y^{14}\sqrt{2x}$ (j) $7x^3\sqrt[3]{3x}$ (k) $18y^5\sqrt[3]{2y}$ (l) $3m^3\sqrt[4]{2m}$. **11.** D.

Lesson 9 Assignment

1. A. **2.** B. **3.** B. **4.** B. **5.** D. **6.** 3 645. **7.** 20. **8.** (a) -9 is an integer and rational: $8 + 4 = 12$ points. (b) $A = 4 + 12 + 18 = 34$, $B = 12 + 4 + 12 = 28$, $C = 12 + 2 + 23 = 37$; **Student** C **wins**.